

CSS scroll-behavior Property Complete Guide to Smooth Scrolling and Navigation

CSS Navigation Mastery

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Chapter 1

Introduction to scroll-behavior

1.1 What is scroll-behavior?

The `scroll-behavior` CSS property specifies how a page should scroll when navigating to anchor links or using JavaScript scroll methods. It enables smooth, animated scrolling instead of instant jumps.

Key facts about scroll-behavior:

- Controls scrolling animation behavior
- Applied to `html` or `body` elements
- Values: `auto` (instant) or `smooth` (animated)
- Works with anchor links (`#hash`)
- Works with JavaScript scroll methods
- Part of CSS Scroll Snap Module
- Widely supported in modern browsers
- Improves user experience with visual feedback

1.2 Why Use scroll-behavior?

Smooth scrolling provides several benefits:

- **Better UX:** Visual continuity when navigating
- **Professional feel:** Smooth animations are more polished
- **Context awareness:** Users see where they're scrolling to
- **Reduced disorientation:** Smooth motion is easier to follow
- **Mobile friendly:** Works on all devices
- **Simple implementation:** Single CSS property
- **Accessible:** Respects reduced motion preferences

1.3 Browser Compatibility

`scroll-behavior` is supported in modern browsers:

- Chrome 61+ (2017)
- Firefox 36+ (2015)
- Safari 15.4+ (2022)
- Edge 79+ (2020)
- Opera 48+ (2017)
- Internet Explorer (not supported)

Chapter 2

Basic Syntax and Values

2.1 scroll-behavior Syntax

```
/* Apply to root element for entire page */
html {
    scroll-behavior: auto;    /* Default - instant scroll */
}

html {
    scroll-behavior: smooth; /* Smooth scrolling animation */
}

/* Or apply to body */
body {
    scroll-behavior: smooth;
}

/* Specific elements with overflow */
.scrollable-container {
    overflow-y: scroll;
    scroll-behavior: smooth;
}

/* Global smooth scrolling */
* {
    scroll-behavior: smooth;
}
```

2.2 Property Values

Value	Description
auto	Instant/abrupt scrolling (default behavior)
smooth	Smooth animated scrolling
inherit	Inherit from parent element
initial	Set to default value
revert	Revert to browser default

2.3 Browser Default Behavior

```
/* Without scroll-behavior (browser default) */
html {
    scroll-behavior: auto; /* Instant scroll to anchor */
}

/* With smooth scroll-behavior */
html {
    scroll-behavior: smooth; /* Animated scroll to anchor */
}

/* Example anchor link behavior */
<!-- Clicking this link: -->
<a href="#section-2">Jump to Section 2</a>

<!-- Without smooth: Instantly scrolls to #section-2 -->
<!-- With smooth: Animates scrolling to #section-2 -->
```

Chapter 3

Practical Implementation

3.1 Page Navigation with Smooth Scrolling

```
<style>
/* Enable smooth scrolling for entire page */
html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    margin: 0;
}

/* Navigation bar */
nav {
    position: fixed;
    top: 0;
    width: 100%;
    background: #333;
    padding: 1rem;
    z-index: 100;
}

nav a {
    color: white;
    text-decoration: none;
    margin-right: 1.5rem;
    transition: color 0.3s ease;
}

nav a:hover {
    color: #0066cc;
}

/* Sections with ID for anchor links */
section {
    padding: 5rem 2rem;
    min-height: 100vh;
}
```

```
        margin-top: 60px;
    }

    section:nth-child(odd) {
        background: #f9f9f9;
    }

    section h2 {
        font-size: 2.5rem;
    }
</style>

<nav>
    <a href="#home">Home</a>
    <a href="#about">About</a>
    <a href="#services">Services</a>
    <a href="#contact">Contact</a>
</nav>

<section id="home">
    <h2>Home</h2>
</section>

<section id="about">
    <h2>About Us</h2>
</section>

<section id="services">
    <h2>Our Services</h2>
</section>

<section id="contact">
    <h2>Contact</h2>
</section>
```

3.2 Table of Contents with Smooth Navigation

```
<style>
    html {
        scroll-behavior: smooth;
    }

    /* Table of contents sidebar */
    .toc {
        position: fixed;
        left: 0;
        top: 0;
        width: 200px;
        height: 100vh;
        background: #f5f5f5;
        padding: 1rem;
        overflow-y: auto;
```



```
}

.toc a {
  display: block;
  padding: 0.5rem;
  color: #333;
  text-decoration: none;
  transition: all 0.3s ease;
  border-left: 3px solid transparent;
}

.toc a:hover,
.toc a.active {
  border-left-color: #0066cc;
  color: #0066cc;
  padding-left: 1rem;
}

/* Main content */
main {
  margin-left: 200px;
  padding: 2rem;
}

h2 {
  margin-top: 3rem;
  padding-top: 1rem;
}
</style>

<aside class="toc">
  <h3>Contents</h3>
  <a href="#introduction">Introduction</a>
  <a href="#basics">Basics</a>
  <a href="#advanced">Advanced</a>
  <a href="#examples">Examples</a>
</aside>

<main>
  <section id="introduction">
    <h2>Introduction</h2>
    <p>Content here...</p>
  </section>

  <section id="basics">
    <h2>Basics</h2>
    <p>Content here...</p>
  </section>
</main>
```

3.3 Back-to-Top Button

```
<style>
  html {
    scroll-behavior: smooth;
  }

  /* Back to top button */
  .back-to-top {
    position: fixed;
    bottom: 2rem;
    right: 2rem;
    padding: 0.75rem 1rem;
    background: #0066cc;
    color: white;
    border: none;
    border-radius: 4px;
    cursor: pointer;
    opacity: 0;
    visibility: hidden;
    transition: all 0.3s ease;
  }

  /* Show button when scrolled down */
  .back-to-top.visible {
    opacity: 1;
    visibility: visible;
  }

  .back-to-top:hover {
    background: #0052a3;
    transform: translateY(-2px);
  }
</style>

<button class="back-to-top" onclick="window.scrollTo({top: 0})">↑
  Back to Top
</button>

<script>
  const backToTop = document.querySelector('.back-to-top');

  window.addEventListener('scroll', () => {
    if (window.scrollY > 300) {
      backToTop.classList.add('visible');
    } else {
      backToTop.classList.remove('visible');
    }
  });
</script>
```

3.4 Scrollable Containers

```
<style>
  /* Smooth scrolling for scrollable container */
  .scrollable {
    height: 300px;
    overflow-y: scroll;
    scroll-behavior: smooth;
    border: 1px solid #ddd;
    padding: 1rem;
  }

  .scrollable::-webkit-scrollbar {
    width: 8px;
  }

  .scrollable::-webkit-scrollbar-track {
    background: #f1f1f1;
  }

  .scrollable::-webkit-scrollbar-thumb {
    background: #888;
    border-radius: 4px;
  }

  .scrollable::-webkit-scrollbar-thumb:hover {
    background: #555;
  }

  .scroll-item {
    padding: 1rem;
    margin: 0.5rem 0;
    background: #f9f9f9;
    border-radius: 4px;
  }
</style>

<div class="scrollable">
  <div class="scroll-item">Item 1</div>
  <div class="scroll-item">Item 2</div>
  <div class="scroll-item">Item 3</div>
  <!-- Many items... -->
</div>
```

Chapter 4

Advanced Techniques

4.1 JavaScript Scroll Methods with scroll-behavior

```
<style>
  html {
    scroll-behavior: smooth;
  }
</style>

<script>
  /* scrollTo() respects scroll-behavior */
  function scrollToElement(elementId) {
    const element = document.getElementById(elementId);
    element.scrollIntoView(); /* Respects smooth scroll-behavior */
  }

  /* scrollIntoView() method */
  document.getElementById('target').scrollIntoView({ behavior: 'smooth'
    });

  /* window.scrollTo() method */
  window.scrollTo({ top: 1000, behavior: 'smooth' });

  /* Smooth scroll with offset */
  function smoothScrollToElement(element, offset = 0) {
    const elementPosition = element.getBoundingClientRect().top +
      window.scrollY;
    window.scrollTo({
      top: elementPosition - offset,
      behavior: 'smooth'
    });
  }

  /* Usage */
  smoothScrollToElement(document.getElementById('section'), 60);
</script>
```

4.2 Conditional Smooth Scrolling

```
/* Smooth scrolling for users who don't prefer reduced motion */
html {
  scroll-behavior: smooth;
}

/* Disable smooth scrolling for users who prefer reduced motion */
@media (prefers-reduced-motion: reduce) {
  html {
    scroll-behavior: auto;
  }
}

/* Enable smooth scrolling only on desktop */
@media (prefers-reduced-motion: no-preference) {
  html {
    scroll-behavior: smooth;
  }
}

/* Smooth scrolling only on large screens */
@media (min-width: 1024px) {
  html {
    scroll-behavior: smooth;
  }
}
```

4.3 Scroll Snap with Smooth Behavior

```
/* Combine scroll-behavior with scroll-snap */
html {
  scroll-behavior: smooth;
}

/* Scroll snap container */
.snap-container {
  scroll-snap-type: y mandatory;
  overflow-y: scroll;
  scroll-behavior: smooth;
  height: 100vh;
}

/* Snap points for sections */
section {
  scroll-snap-align: start;
  scroll-snap-stop: always;
  height: 100vh;
}

/* Smooth scrolling between snap points */
```

4.4 Progressive Enhancement

```
/* Feature detection with @supports */
@supports (scroll-behavior: smooth) {
  html {
    scroll-behavior: smooth;
  }
}

/* Fallback without smooth scrolling */
@supports not (scroll-behavior: smooth) {
  html {
    scroll-behavior: auto;
  }
}
```

Chapter 5

Real-World Examples

5.1 Example 1: Documentation Site

```
<style>
  html {
    scroll-behavior: smooth;
  }

  body {
    display: grid;
    grid-template-columns: 250px 1fr;
    gap: 0;
  }

  /* Sidebar navigation */
  .sidebar {
    background: #f5f5f5;
    padding: 1.5rem;
    position: sticky;
    top: 0;
    height: 100vh;
    overflow-y: auto;
  }

  .sidebar a {
    display: block;
    padding: 0.5rem 1rem;
    color: #333;
    text-decoration: none;
    border-left: 3px solid transparent;
    transition: all 0.3s ease;
  }

  .sidebar a:hover,
  .sidebar a.active {
    border-left-color: #0066cc;
    color: #0066cc;
  }
}
```

```
/* Main content */
main {
  padding: 2rem;
}

section {
  margin: 2rem 0;
  padding: 1rem 0;
  border-bottom: 1px solid #eee;
}

section:first-child {
  border-top: 1px solid #eee;
}
</style>

<div class="sidebar">
  <h3>Documentation</h3>
  <a href="#getting-started">Getting Started</a>
  <a href="#installation">Installation</a>
  <a href="#usage">Usage</a>
  <a href="#api">API Reference</a>
</div>

<main>
  <section id="getting-started">
    <h2>Getting Started</h2>
    <p>Documentation content...</p>
  </section>
  <section id="installation">
    <h2>Installation</h2>
    <p>Installation steps...</p>
  </section>
</main>
```

5.2 Example 2: Single Page Portfolio

```
<style>
  html {
    scroll-behavior: smooth;
  }

  /* Fixed navigation */
  nav {
    position: fixed;
    top: 0;
    width: 100%;
    background: rgba(0, 0, 0, 0.9);
    padding: 1rem 2rem;
    z-index: 100;
  }
```



```
nav a {
  color: white;
  text-decoration: none;
  margin: 0 1.5rem;
  transition: color 0.3s ease;
}

nav a:hover {
  color: #0066cc;
}

/* Full-screen sections */
section {
  min-height: 100vh;
  display: flex;
  align-items: center;
  justify-content: center;
  margin-top: 60px;
  padding: 2rem;
  text-align: center;
}

section h2 {
  font-size: 3rem;
  margin: 0;
}

section:nth-child(even) {
  background: #f9f9f9;
}
</style>

<nav>
  <a href="#portfolio">Portfolio</a>
  <a href="#about">About</a>
  <a href="#services">Services</a>
  <a href="#contact">Contact</a>
</nav>

<section id="portfolio">
  <h2>My Work</h2>
</section>

<section id="about">
  <h2>About Me</h2>
</section>

<section id="services">
  <h2>Services</h2>
</section>

<section id="contact">
  <h2>Contact</h2>
</section>
```

5.3 Example 3: Blog with Smooth Scrolling

```
<style>
  html {
    scroll-behavior: smooth;
  }

  /* Article layout */
  article {
    max-width: 800px;
    margin: 0 auto;
    padding: 2rem;
  }

  /* Table of contents */
  .toc {
    background: #f5f5f5;
    padding: 1.5rem;
    border-radius: 8px;
    margin: 2rem 0;
  }

  .toc h3 {
    margin-top: 0;
  }

  .toc ul {
    list-style: none;
    padding: 0;
  }

  .toc a {
    color: #0066cc;
    text-decoration: none;
    transition: color 0.3s ease;
  }

  .toc a:hover {
    color: #0052a3;
  }

  /* Article sections */
  article section {
    margin: 3rem 0;
  }

  article h2 {
    border-bottom: 2px solid #0066cc;
    padding-bottom: 0.5rem;
  }
```

```
</style>

<article>
  <h1>Article Title</h1>

  <div class="toc">
    <h3>Contents</h3>
    <ul>
      <li><a href="#introduction">Introduction</a></li>
      <li><a href="#main-topic">Main Topic</a></li>
      <li><a href="#conclusion">Conclusion</a></li>
    </ul>
  </div>

  <section id="introduction">
    <h2>Introduction</h2>
    <p>Article content...</p>
  </section>

  <section id="main-topic">
    <h2>Main Topic</h2>
    <p>Article content...</p>
  </section>

  <section id="conclusion">
    <h2>Conclusion</h2>
    <p>Article content...</p>
  </section>
</article>
```

Chapter 6

Best Practices and Accessibility

6.1 Accessibility Considerations

```
/* Respect user's motion preferences */
@media (prefers-reduced-motion: reduce) {
  html {
    scroll-behavior: auto;
  }

  * {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }
}

/* Enable smooth scroll for users who prefer it */
@media (prefers-reduced-motion: no-preference) {
  html {
    scroll-behavior: smooth;
  }
}

/* Ensure focus is visible */
a:focus {
  outline: 2px solid #0066cc;
  outline-offset: 2px;
}

/* Skip to content link */
.skip-to-content {
  position: absolute;
  top: -40px;
  left: 0;
  background: #0066cc;
  color: white;
  padding: 0.5rem 1rem;
  text-decoration: none;
}
```

```
.skip-to-content:focus {  
  top: 0;  
}
```

6.2 Performance Tips

- **Minimal impact:** scroll-behavior has negligible performance cost
- **Use sparingly:** Limit the number of scroll triggers
- **Combine with other effects:** Use with animations carefully
- **Test on mobile:** Smooth scrolling can feel different on touch devices
- **Consider duration:** Very long scrolls may feel sluggish

6.3 Best Practices

1. **Respect user preferences:** Check `prefers-reduced-motion`
2. **Use meaningful anchor links:** Ensure IDs match navigation targets
3. **Provide visual feedback:** Highlight active navigation items
4. **Test thoroughly:** Verify on different browsers and devices
5. **Combine with other techniques:** Use with scroll-snap for better UX
6. **Keep scroll distance reasonable:** Very long pages may need infinite scroll
7. **Document anchor links:** Make navigation obvious to users

Chapter 7

Quick Reference

7.1 scroll-behavior Syntax

```
/* Basic syntax */
html {
    scroll-behavior: smooth;
}

/* Auto scrolling (default) */
html {
    scroll-behavior: auto;
}

/* For scrollable containers */
.container {
    overflow-y: scroll;
    scroll-behavior: smooth;
}

/* Respecting user preferences */
@media (prefers-reduced-motion: reduce) {
    html {
        scroll-behavior: auto;
    }
}

@media (prefers-reduced-motion: no-preference) {
    html {
        scroll-behavior: smooth;
    }
}
```

7.2 Common Implementation Patterns

```
/* Pattern 1: Global smooth scrolling */
html {
    scroll-behavior: smooth;
}
```

```
}

/* Pattern 2: Smooth scrolling with reduced motion support */
html {
  scroll-behavior: smooth;
}

@media (prefers-reduced-motion: reduce) {
  html {
    scroll-behavior: auto;
  }
}

/* Pattern 3: Smooth scrolling for specific containers */
.scrollable-area {
  overflow-y: scroll;
  scroll-behavior: smooth;
  height: 400px;
}

/* Pattern 4: Smooth scroll with scroll snap */
html {
  scroll-behavior: smooth;
  scroll-snap-type: y mandatory;
}

section {
  scroll-snap-align: start;
}
```

7.3 Production-Ready Template

```
/* Complete scroll-behavior setup */

/* Feature detection */
html {
  scroll-behavior: smooth;
}

/* Respect user motion preferences */
@media (prefers-reduced-motion: reduce) {
  html {
    scroll-behavior: auto;
  }

  * {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }
}
```

```
/* Scrollable containers */
.scrollable {
  overflow-y: scroll;
  scroll-behavior: smooth;
}

/* Skip to content link */
.skip-link {
  position: absolute;
  top: -40px;
  left: 0;
  background: #0066cc;
  color: white;
  padding: 0.5rem 1rem;
  text-decoration: none;
}

.skip-link:focus {
  top: 0;
  z-index: 100;
}

/* Anchor link styling */
[id] {
  scroll-margin-top: 60px; /* Offset for fixed headers */
}

/* Navigation active state */
nav a[href$="#active"] {
  color: #0066cc;
  border-bottom: 2px solid #0066cc;
}
```

7.4 Browser Support Fallback

```
<script>
  // Check if scroll-behavior is supported
  function supportsScrollBehavior() {
    return 'scrollBehavior' in document.documentElement.style;
  }

  if (!supportsScrollBehavior()) {
    // Fallback for browsers without scroll-behavior support
    document.addEventListener('click', (e) => {
      if (e.target.tagName === 'A' && e.target.hash) {
        e.preventDefault();
        const target = document.querySelector(e.target.hash);
        if (target) {
          // Polyfill smooth scroll
          smoothScroll(target);
        }
      }
    })
  }
}
```



```
    });  
  }  
  
  function smoothScroll(element) {  
    const startTop = window.scrollY;  
    const elementTop = element.getBoundingClientRect().top +  
      window.scrollY;  
    const distance = elementTop - startTop;  
    const duration = 1000;  
    const start = Date.now();  
  
    const animate = () => {  
      const now = Date.now();  
      const progress = (now - start) / duration;  
      const ease = progress < 0.5  
        ? 2 * progress * progress  
        : -1 + (4 - 2 * progress) * progress;  
  
      window.scrollTo(0, startTop + distance * ease);  
  
      if (progress < 1) {  
        requestAnimationFrame(animate);  
      }  
    };  
  
    requestAnimationFrame(animate);  
  }  
</script>
```